

The Blueprint Concept

In Java, a **Class** serves as the blueprint, while an **Object** is the real-world instance created from that template. Think of a Car design (Class) vs. the actual Car on the road (Object).

 Remember: Class names must always start with an Uppercase letter, and the filename must match exactly.

Creation & The 'new' Keyword

A class defines variables (attributes) and methods (behaviors). To use them, you must instantiate the class using the new keyword.

```
public class Main {  
    int x = 5; // Attribute  
  
    public static void main(String[] args) {  
        Main myObj = new Main(); // Creation  
        System.out.println(myObj.x);  
    }  
}
```

Multi-Object Instances

Multiple objects can be created from a single class. Each object maintains its own unique state (data).

```
Main obj1 = new Main();
Main obj2 = new Main();

obj1.x = 10;
obj2.x = 20; // Independent of obj1
```

Class vs. Object Summary

Feature	Class	Object
Definition	The Blueprint / Template	The Real Instance
Existence	Declared once in code	Created multiple times at runtime
Memory	Stored in Method Area	Stored in Heap Memory

Multi-File Architecture

Professionally, Java programs are split into multiple files: one to define the data structures and another (Main) to execute logic.

- `DataClass.java`: Holds attributes & methods.
- `AppRunner.java`: Contains the `main()` method.